Folded homoclinic bifurcations as a route to spiking and bursting

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Abstract. Abstract to be written

Key words. spiking, bursting, folded saddle, homoclinic bifurcation, slow-fast

MSC codes. 34C26, 34E13, 34E17, 37N25

1. Introduction.

Complex oscillations, including the phenomena known as spiking and bursting, have been of great interest to physiologists and mathematical biologists for some time. Loosely speaking, we say that a *spike* occurs when the time series for a solution exhibits a rapid increase in amplitude of one or more variables, followed almost immediately by a rapid fall in amplitude, and that a *burst* occurs when a time series shows a number of spikes in quick succession followed by a period of relative quiescence. Figure 1 shows examples of spikes and bursts in experimental data from L5 cortical neurons of Wistar rat showing spiking (panel (a)) and bursting ((b)) responses to a step current. Spikes and bursts occur in many physiological contexts, perhaps most notably in neural dynamics and intracellular calcium dynamics [2].

Mathematical models of systems with spiking and bursting commonly have multiple timescales, reflecting the different timescales on which processes occur in the underlying physiological systems. Early work on models of spiking and bursting focussed particularly on systems with just one slow variable, and classified the type of bursting by characteristics of the fast dynamics (see, for example, [11], [1], [7], [6]). Some studies have been devoted to understanding the *onset* of spiking and bursting, which occurs when a system parameter is varied and the system or model undergoes a transition from having an attracting equilibrium solution or small amplitude oscillation to exhibiting spiking or bursting solutions [5]. Again, the focus in early work was mostly on transitions in systems with one slow variable.

More recent work has extended the classification of spiking and bursting by investigating these phenomena in systems in which there are two slow variables. An overview of some of the relevant work is provided in [5]. Of particular interest here is [4], which identified a specific mechanism, a so-called *folded homoclinic bifurcation*, for the onset of spiking and bursting in several systems with two slow variables. One of the systems studied in [4] was a four-dimensional polynomial model developed as a simplified version of the Plant model for the Aplysia R15 bursting neuron [10], a conductance-based neural model. The equations for that model are given in Appendix A.1 with a bifurcation diagram and some selected time series shown in Figure 2. It can be seen that a branch of periodic orbits is generated in a supercritical Hopf bifurcation at $k \approx 0.97$. The oscillations start out being of small amplitude but, as k increases, the branch of periodic orbits has a series of 'barbs' and passage through

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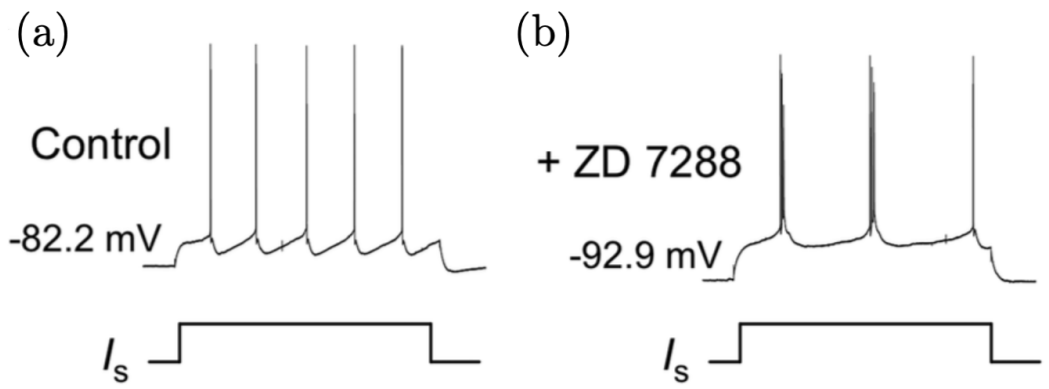


Figure 1. *Spiking and bursting in L5 layer of the cortex of Wistar rats; modified from [9].*

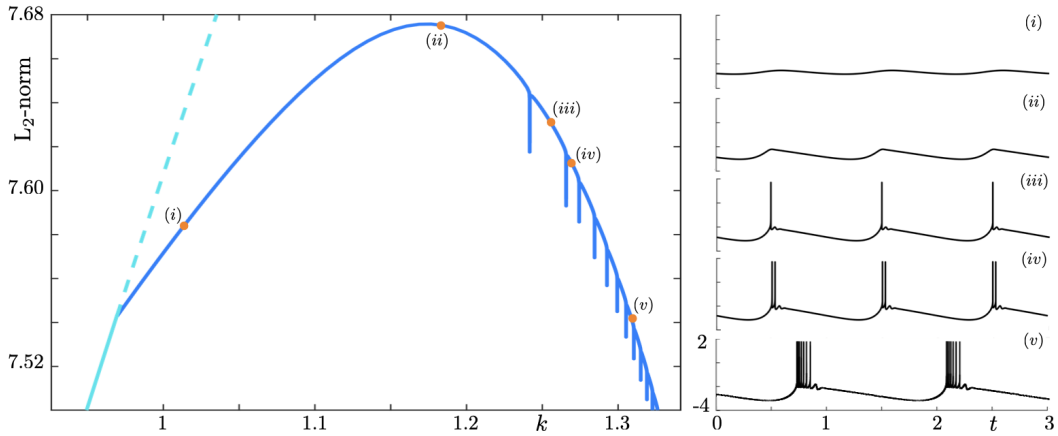


Figure 2. *Bifurcation diagram and selected time series for equations (A.1), with parameter values as in Table 1. The left panel shows the bifurcation diagram; the cyan curve indicates an equilibrium solution (solid for stable, dashed for unstable) and the blue curve shows the L_2 norm for solutions on a branch of stable periodic orbits. The right panel shows time series for the variable x for solutions corresponding to the labelled orange dots on the branch of periodic solutions in the bifurcation diagram.*

each barb results in a spike being added to the periodic orbit.

Desroches et al. [4] used GSPT to show that the onset of the first spike occurred at a parameter value close to that for which there was a homoclinic bifurcation of a saddle equilibrium in the DRS with this saddle corresponding to a FS in the reduced system. We (follow [4] and) call this bifurcation in the reduced problem a *folded homoclinic bifurcation* (FHB). Talk about Fig. 3. While there is no corresponding bifurcation in the full system, equations (A.1), there is a clear change in the nature of the attracting periodic solution (the appearance of a spike) associated with the bifurcation in the reduced problem. Desroches et al. [4] made this observation and hypothesised that a FHB might be a mechanism for the onset of spiking in other models (including BRC which was investigated a bit in this paper), but did not undertake a general analysis of FHBs. The purpose of this article is to dig into it more, to try to understand in more detail the geometric mechanisms that may lead of spiking

and bursting in models with an FHB.

We will show that complex dynamics may arise from FHB (or, more accurately, a series of FHBs), with the details depending critically on the relative positions of the branches of stable and unstable manifolds of the FS on the attracting sheet of the critical manifold (need to define CM) of of the projections onto the attracting sheet of the branches of these manifolds lying on the nearby unstable sheet. For some configurations of these manifolds it turns out that no spiking is induced by the FHB, as will be discussed later.

The branch of periodic orbits shown in Figure 2 is continuous, that is, each spike arises in a continuous transition as k passes through the corresponding barb. It is not always the case that spike adding is a continuous process, as the example in Figure 4 shows; this example was also in the [4] paper. In that figure, the red branch of periodic solutions originates in a supercritical Hopf bifurcation at $I \approx -4.35$, undergoes a fold at $I \approx -4.08$, grows in amplitude without developing a clear spike as I decreases and terminates in a homoclinic bifurcation at $I \approx -4.11$. A new branch of attracting periodic orbits is created in a homoclinic bifurcation at $I \approx -4.11$; as I increases, the periodic orbits develops a clear spike. The branch undergoes a fold at $I \approx -3.9$ then terminates in another periodic orbit at $I \approx -3.96$. Figure 4 shows five branches of periodic orbits in total, with all but the red branch having essentially the same shape.

In the remainder of the paper, we use geometric arguments to understand the dynamics that may be associated with FHB bifurcations, then show which of the different cases arise in the two models discussed above. We look at the onset of the first spike-adding transition (which is essentially a local process) in section 2. Once spike-adding has been initiated, the development of a full spike is controlled by global geometry of the system, and we look at two distinct cases in section 3. We also show how bursting develops through a series of transitions that are very similar to the development of the first spike. Our conclusions follow in section 4. In the following, we assume a basic understanding of the methods of geometric singular perturbation theory (GSPT) for the analysis of systems with multiple timescales and refer interested readers to [8] for more details on GSPT.

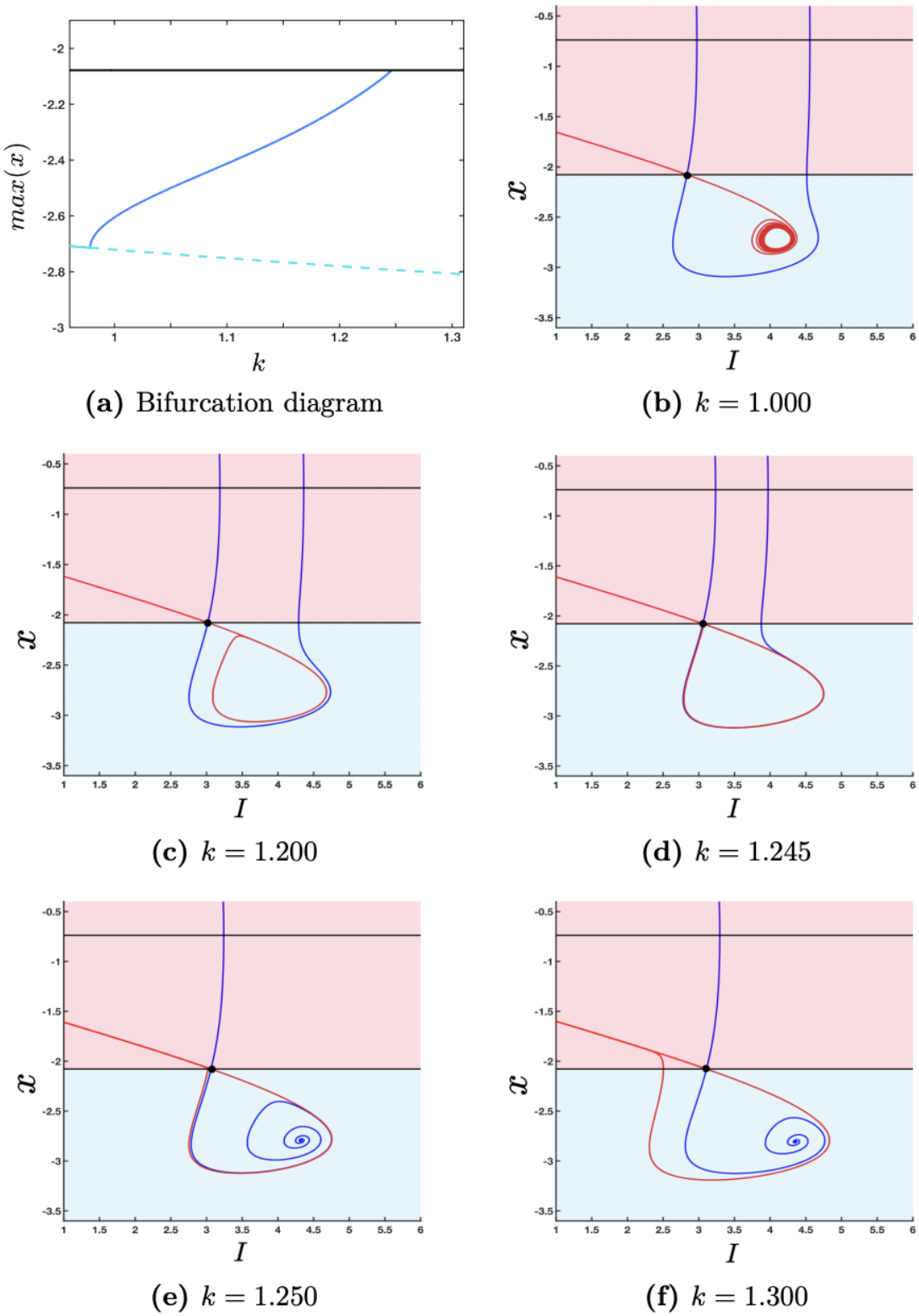
2. Spike initiation.

Define a droplet orbit? Focus in this section is on the geometric characteristics of droplet orbits and the circumstances under which these are precursors to the development of full spikes and then bursting.

We start by considering the simplest setting in which a homoclinic bifurcation involving a folded saddle in the DRS might occur: a three-dimensional two-timescale system with one fast and two slow variables,

$$\begin{aligned} \frac{dx}{dt} &= f(x, y_1, y_2, \varepsilon, \nu), \\ \frac{dy_1}{dt} &= \varepsilon g_1(x, y_1, y_2, \varepsilon, \nu), \\ \frac{dy_2}{dt} &= \varepsilon g_2(x, y_1, y_2, \varepsilon, \nu), \end{aligned} \tag{2.1}$$

where $(x, y_1, y_2) \in \mathbb{R}^3$, the functions f, g_1, and g_2 are sufficiently smooth and $\mathcal{O}(1)$, $0 \leq \varepsilon \ll 1$ and ν is a bifurcation parameter. These equations give the system on the fast timescale, t.



(a) Bifurcation diagram (b) $k = 1.000$ (c) $k = 1.200$ (d) $k = 1.245$ (e) $k = 1.250$ (f) $k = 1.300$

Figure 3. *This is a screenshot. Bifurcation diagram and phase portraits of the DRS for Plant-Based Model, equations (A.1)), with parameter values as in Table 1. (a) Bifurcation diagram. The black curve shows a saddle equilibrium on the lower fold of S. The cyan curve indicates a regular equilibrium on the lower sheet of S (solid for stable, dashed for unstable). The blue curve shows the maximum value of x of stable periodic orbits on the lower sheet of S.*

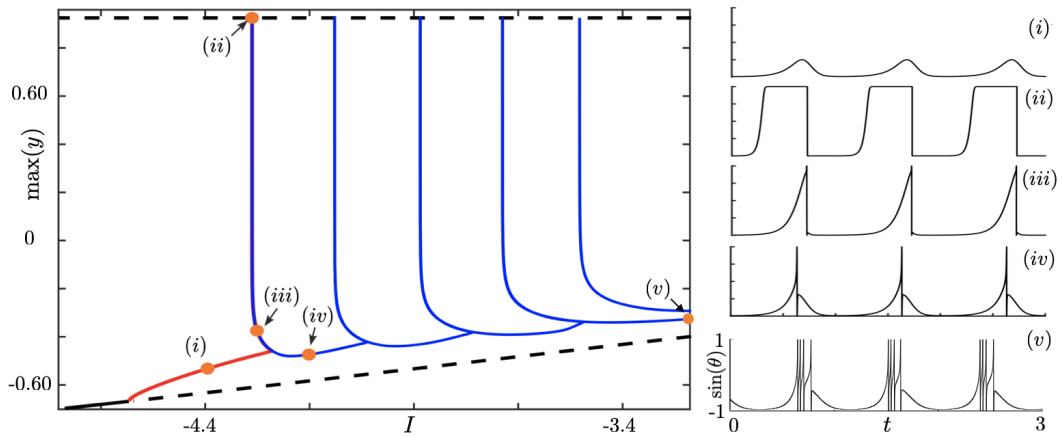


Figure 4. *Bifurcation diagram and time series for the BRC system, equations (A.7), with parameter values as in Table 2. The left panel shows the bifurcation diagram; the cyan curves indicate equilibrium points (solid for stable, dashed for unstable) while the red and blue curves show the maximum y value for solutions on various branches of periodic orbits. The right panels show tIme series for the solutions corresponding to the labelled orange dots on the branch of periodic solutions in the bifurcation diagram.*

Rescaling time using $t = \varepsilon\tau$ gives the system on the slow timescale,

$$\begin{aligned} \varepsilon\frac{dx}{d\tau} &= f(x, y_1, y_2, \varepsilon, \nu), \\ \frac{dy_1}{d\tau} &= g_1(x, y_1, y_2, \varepsilon, \nu), \\ \frac{dy_2}{d\tau} &= g_2(x, y_1, y_2, \varepsilon, \nu). \end{aligned} \tag{2.2}$$

Taking $\varepsilon \to 0$ in system (2.1) yields the layer problem,

$$\begin{aligned} \frac{dx}{dt} &= f(x, y_1, y_2, \varepsilon, \nu), \\ \frac{dy_1}{dt} &= 0, \\ \frac{dy_2}{dt} &= 0, \end{aligned} \tag{2.3}$$

and taking $\varepsilon \to 0$ in system (2.2) yields the reduced problem,

$$\begin{aligned} 0 &= f(x, y_1, y_2, 0, \nu), \\ \frac{dy_1}{d\tau} &= g_1(x, y_1, y_2, 0, \nu), \\ \frac{dy_2}{d\tau} &= g_2(x, y_1, y_2, 0, \nu). \end{aligned} \tag{2.4}$$

Equilibria of the layer problem make up the critical manifold,

$$S := \{(x, y_1, y_2) \mid f(x, y_1, y_2, 0, \nu) = 0\}.$$

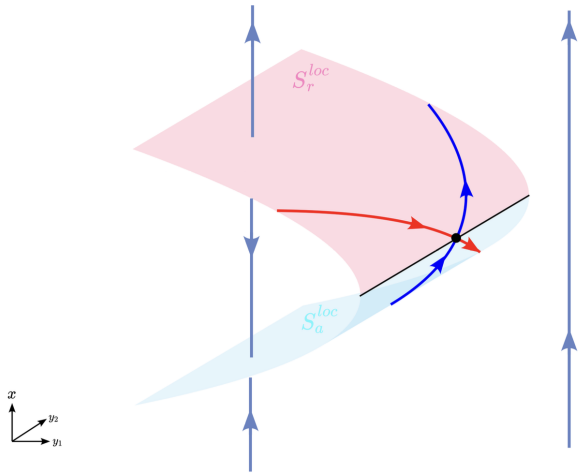


Figure 5. *The critical manifold S^{loc} for the local system (2.1). The stable portion of S^{loc}, S_a^{loc}, is shown by the cyan-shaded sheet and the unstable portion of S^{loc}, S_r^{loc}, is shown by the pink-shaded sheet. Fold points are shown by the black line. Fast fibres of the layer problem (2.3) are shown as vertical light blue lines; arrows indicate direction of flow. A folded saddle of (2.6) (black dot) sits on the fold line. The blue and red curves on the critical manifold show the true and faux canards of (2.4), respectively; arrows indicate the direction of flow.*

We assume a portion of the critical manifold is a nondegenerate folded surface made up of three components, $S^{loc} = S_a^{loc} \cup L^{loc} \cup S_r^{loc}$: a sheet of attracting equilibria,

$$S_a^{loc} := \{(x, y_1, y_2) \in S^{loc} \mid f_x(x, y_1, y_2, 0, \nu) < 0\},$$

a sheet of repelling equilibria,

$$S_r^{loc} := \{(x, y_1, y_2) \in S^{loc} \mid f_x(x, y_1, y_2, 0, \nu) > 0\}$$

and a line of fold points,

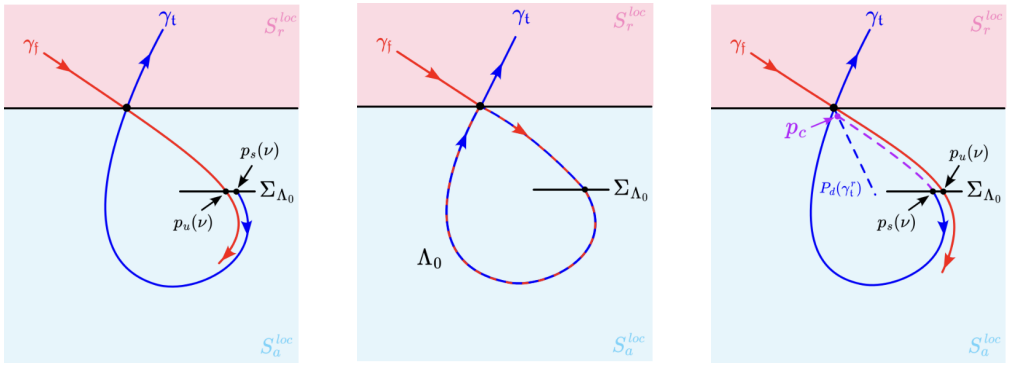
$$L^{loc} := \{(x, y_1, y_2) \in S^{loc} \mid f_x(x, y_1, y_2, 0, \nu) = 0\}$$

separating the two sheets. This is illustrated in Figure 5.

The layer problem (2.3) describes the dynamics on the set of one-dimensional fast fibres, which are parallel to the x-axis. Each point on S_r^{loc} has an associated fast fibre that forms a heteroclinic connection to the point on S_a^{loc} with the same values of y_1 and y_2, as shown in Figure 5. Expressing S^{loc} as a graph over x and y_2 gives $y_1 = \phi(x, y_2)$, which can be used to project (2.4) onto the (x, y_2)-space. As explained in, e.g., [3], it is then possible to obtain

$$\begin{aligned} -\frac{df}{dx}\frac{dx}{d\tau} &= \frac{df}{dy_1}g_1 + \frac{df}{dy_2}g_2, \\ \frac{dy_2}{d\tau} &= g_2. \end{aligned} \tag{2.5}$$

Rescaling time using the change of coordinates $\tau = -\frac{df}{dx}\bar{\tau}$ leads to the DRS



(a) $\nu < 0$ (b) $\nu = 0$ (c) $\nu > 0$

Figure 6. *Projection onto the $x - y_2$ plane of the reduced dynamics (2.4) near a FHB. Colours as in Fig. 5. $p_s(\nu)$ and $p_u(\nu)$ are the horizontal component (y_2-coordinate) of the point of intersection of γ_t and γ_f with the section Σ_{Λ_0}, respectively. On S_a^{loc}, γ_t and γ_f undergo a homoclinic bifurcation. (a) γ_t lies outside γ_f when $p_s(\nu) > p_u(\nu)$. (b) A homoclinic connection, Λ_0, occurs when $p_s(\nu) = p_u(\nu)$. (c) γ_f lies outside γ_t when $p_s(\nu) < p_u(\nu)$. $P_d(\gamma_t^r)$ (blue-dashed line) is the portion of γ_t on S_r^{loc} projected onto S_a^{loc} following fast fibres of the layer problem. The purple-dashed line shows the backwards flow of the reduced problem from $p_s(\nu)$, which intersects $P_d(\gamma_t^r)$ at p_c.*

$$\begin{aligned} \frac{dx}{d\bar{\tau}} &= \frac{df}{dy_1} g_1 + \frac{df}{dy_2} g_2, \\ \frac{dy_2}{d\bar{\tau}} &= -\frac{df}{dx} g_2. \end{aligned} \tag{2.6}$$

Assuming $f_{y_1} g_1 + f_{y_2} g_2 = 0$ at some point $l_{\text{fs}} \in L^{loc}$ and $g_2(l_{\text{fs}}; 0, \nu) \neq 0$, it follows that l_{fs} is a folded singularity in (2.6). Let λ_1 and λ_2 denote the eigenvalues of (2.6) linearised about l_{fs}, and assume $\lambda_1 \lambda_2 < 0$. Then l_{fs} is of saddle type. We further assume that the saddle index, $\sigma = \lambda_1 + \lambda_2 \neq 0$.

We assume the DRS (system (2.6)) has a homoclinic bifurcation involving l_{fs} when $\nu = 0$, with a corresponding FHB in the reduced problem (2.5). Denote the homoclinic orbit by Λ_0 and assume that it exists on the attracting sheet of S^{loc}; see Figure 6 (b). We assume that as ν is varied from 0 the branches of the stable and unstable manifolds of l_{fs} on S_a^{loc} split as shown in Figure 6. Specifically, near $\nu = 0$, we introduce a one-dimensional local cross-section, Σ_{Λ_0}, transverse to the unstable manifold of l_{fs} on S_a^{loc} near l_{fs} and define $p_s(\nu)$ and $p_u(\nu)$ as the value of the horizontal component (y_2-coordinate) of the first intersection point of the stable and unstable manifolds of l_{fs} with Σ_{Λ_0}, respectively; see Figure 6 (a). We define the splitting distance as $p(\nu) = p_u(\nu) - p_s(\nu)$, with the convention that $p(\nu) < 0$ if $\nu < 0$ and $p(\nu) > 0$ if $\nu > 0$ and assume that $p'(0) \neq 0$; see Figure 6.

Ensure to include

- The three cases, all with negative saddle index: need to think more carefully about which of the many possibilities to include. Discuss case where there is no spiking resulting from FHB

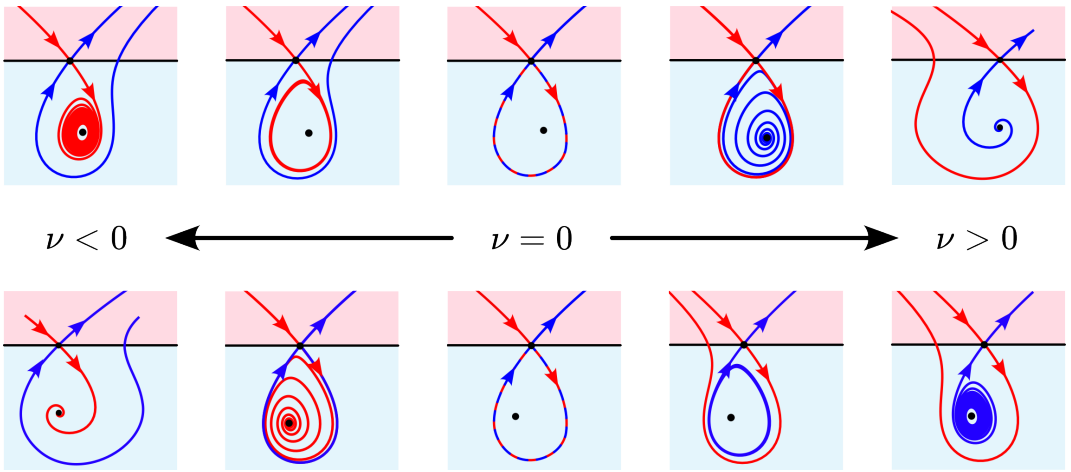


Figure 7. *Geometry of the folded saddle's manifolds in (2.4) for various values of ν. S_a^{loc} is shown by the cyan-shaded sheet, S_r^{loc} is shown by the pink-shaded sheet and a line of fold points, L^{loc}, shown in black, separates them. Along the line of fold points is a folded saddle (black dot). Shown in blue is γ_t and shown in red is γ_f. A regular equilibrium point exists on S_a^{loc} as shown by the black dot. The top row is the case for a negative saddle index and the bottom row is the case for a positive saddle index. In both sequences the folded homoclinic connection occurs at $\nu = 0$.*

- Some related bifurcation diagrams from the two models that illustrate the points?

3. Fully developed spiking and bursting.

- TODO
- Some text about averaged reduced problems

4. Conclusions.

- A periodic orbit lies near an attracting sheet of the critical manifold. The periodic orbit grows in amplitude until it approaches a fold curve of the critical manifold.
- As the periodic orbit approaches the fold curve, part of it leaves the attracting sheet and passes over to the repelling sheet, developing a small canard segment. Small oscillations near the location of the folded saddle may be observed. The canard segment is followed by a fast jump back towards the attracting sheet and then the periodic orbit closes. Periodic orbits containing a canard segment followed by a fast jump back segment will be referred to as *droplet orbits* in the following.
- The canard segment continues to grow in size along the repelling sheet until the maximal canard is reached (PB model, BRC model, TTS model) or some bifurcation terminates the branch (Minimal model, TTS model).
- In cases where a maximal canard is reached, a periodic orbit continues to exist when the bifurcation parameter is varied further but the nature of the periodic orbit changes. Specifically, the canard segment begins to shrink again and the fast jump following the canard segment is now upwards rather than back to the original sheet. Periodic orbits containing upward fast jumps after the canard segment will be referred to as *spikelet orbits* in the following. Eventually, the canard segment disappears completely and a fully developed spike emerges.

- Quick summary of what an FHB is.
- The FHB is an interesting object, particularly as a route to spiking and bursting, and arises in a number of models.
- The name of the object is misleading - not a bifurcation and the complex dynamics (when it occurs) arises from something other than the existence of the homoclinic bifurcation in the DRS. In particular, saddle index is often regarded as an important quantity for understanding homoclinic bifurcations, but it turns out to be more or less irrelevant for onset of spiking and bursting other than overall stability of the initial slow periodic orbit.
- Overlooked until recently as a mechanism for onset of bursting and spiking. Also, folded things are commonly looked at but less so folded saddle.
- Clear link with parabolic bursting. Or is it triangular bursting? May want to discuss the relationship of PB model to cod2 SNHom point. Perhaps adjusting parameters in PB model could lead to pure parabolic bursting?
- Mention that we have restricted to cases where the slow periodic orbit lies entirely on one sheet of CM (attracting sheet in our case). Others (e.g., Guckenheimer) have looked at cases where the periodic orbit involved in the homoclinic bifurcation lies on more than one sheet.
- Summary of dynamics seen, including mention of cases where there is no spiking or bursting arising from the FHB.
- Return maps not helpful and explain why.
- Other models where FHB might be seen (including two more from Morgan if these have not already been discussed). Also, mention the propofol example of Martin and Nancy.
- There should be a link to SNICeroclinic work of Krasi, Pete and student, at least in the sense that there is more to be understood about how slow flow can be important.

Appendix A. Model equations and parameters.

A.1. Plant-based model.

The equations

$$\begin{aligned}
\frac{dx}{dt} &= c\left(x - x^3/3 - y + z + I\right) \equiv f_1(x, y, z, I, \varepsilon),\\
\frac{dy}{dt} &= \left(x^2 + dx - by + a\right)/c \equiv f_2(x, y, \varepsilon),\\
\frac{dz}{dt} &= \varepsilon\left(z - s(x - x_1)\right) \equiv \varepsilon g_1(x, z, \varepsilon),\\
\frac{dI}{dt} &= \varepsilon\left(k - h_x(x - x_f)^2 - h_y(y - y_f)^2 - h_I(I - I_f)\right) \equiv \varepsilon g_2(x, y, I, \varepsilon)
\end{aligned} \tag{A.1}$$

are referred to in this article as the *Plant-based model*. Here, t denotes the fast timescale and $0 < \varepsilon \ll 1$.

In the numerical computations presented, we use these equations with default values of parameters specified in Table 1. The parameters x_f, y_f and I_f are chosen to assure certain

Table 1
Default parameter values used in the simulations of the Plant-based model (A.1)

$a = 0.08$	$b = 0.71$	$c = 3.00$	$d = 1.80$	$\varepsilon = 0.0004$
$s = 4.00$	$x_1 = -1.6$	$h_x = 0.1$	$h_y = 0.1$	$h_I = 0.1$

desirable geometric properties of the system, as in [4]:

$$\begin{aligned} x_f = -\frac{1}{b}\left(1+\sqrt{1+b(b-d)}\right), \quad y_f = \frac{1}{b}\left(x_f^2 - dx_f + a\right), \\ I_f = \frac{1}{3}x_f^3 + \frac{1}{b}(x_f^2 + dx_f + a) - x_f - s(x_f - x_1). \end{aligned} \tag{A.2}$$

The quantity k is treated as a bifurcation parameter.

Following standard procedure from geometric singular perturbation theory, we introduce the slow timescale $\tau = \varepsilon t$ and rewrite (A.1) as

$$\begin{aligned} \varepsilon\frac{dx}{d\tau} &= f_1(x,y,z,I,\varepsilon), \\ \varepsilon\frac{dy}{d\tau} &= f_2(x,y,\varepsilon), \\ \frac{dz}{d\tau} &= g_1(x,z,\varepsilon), \\ \frac{dI}{d\tau} &= g_2(x,y,I,\varepsilon) \end{aligned} \tag{A.3}$$

Taking the singular limit, $\varepsilon \to 0$, of (A.1) yields the *layer problem*

$$\begin{aligned} \frac{dx}{dt} &= f_1(x,y,z,I,0), \\ \frac{dy}{dt} &= f_2(x,y,0), \\ \frac{dz}{dt} &= 0, \\ \frac{dI}{dt} &= 0. \end{aligned} \tag{A.4}$$

Taking the singular limit, $\varepsilon \to 0$, of (A.3) yields the *reduced problem*

$$\begin{aligned} 0 &= f_1(x,y,z,I,0), \\ 0 &= f_2(x,y,0), \\ \frac{dz}{d\tau} &= g_1(x,z,0), \\ \frac{dI}{d\tau} &= g_2(x,y,I,0). \end{aligned} \tag{A.5}$$

The set of equilibria of (A.4),

$$S := \{(x,y,z,I) \mid f_1(x,y,z,I,0) = 0 \text{ and } f_2(x,y,0) = 0\}$$

forms the *critical manifold*. Writing S as a graph of z over x and I yields

$$z = x^3/3 + x^2/b + (d/b - 1)x + a/b - I. \tag{A.6}$$

Plotted with respect to (x, y, I) coordinates, S consists of three two-dimensional sheets separated by two folds defined by $dz/dx = 0$ and located at $x_f^{\pm} = -\frac{1}{b}(1 \pm \sqrt{1 + b(b-d)})$.

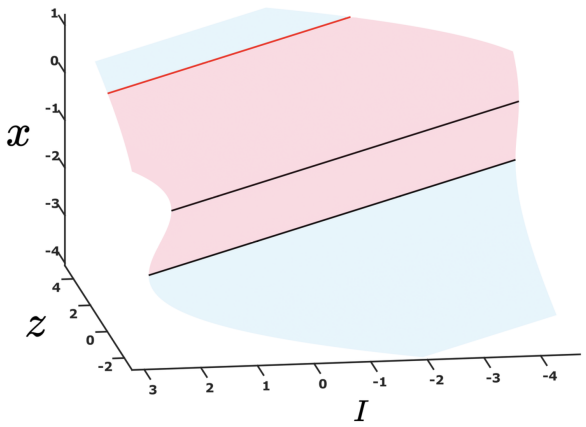


Figure 8. *The critical manifold for the Plant-based model, equations (A.1), with parameter values as in Table 1 and equations (A.2). Stable equilibria of (A.4) are shown by the cyan-shaded sheets and unstable equilibria of (A.4) are shown by the pink-shaded sheets. The folds of the critical manifold relative to z are shown by the black lines. The red line marks the separation of the stable and unstable sections of the upper sheet; if z and I are regarded as parameters of (A.4) then the red line is a curve of Hopf bifurcations of (A.4).*

Figure 8 shows the critical manifold in (x, z, I)-space. The lower sheet consists of stable equilibrium points of (A.4). The middle sheet consists of equilibria of (A.4) that are of saddle type. The upper sheet of equilibria of (A.4) is divided by the red line: between the upper fold and the red line the equilibria are unstable and on the other side of the red line the equilibria are stable within the layer problem.

Need to add equations for DRS?

A.2. BRC model. The equations

$$\begin{aligned} \frac{d\theta}{dt} &= 1 - \cos\theta + A(x, y), \\ \frac{dx}{dt} &= \varepsilon k \left(\sin(p_x + \theta) - x\right), \\ \frac{dy}{dt} &= \varepsilon \left(\sin(p_y + \theta) - y\right), \end{aligned} \tag{A.7}$$

where $A(x, y) = \tanh(ax - by + I)$, are referred to in this article as the BRC Model. Here t denotes the fast timescale and $0 < \varepsilon \ll 1$.

In the numerical computations presented, we use these equations with default values of parameters as in Table 2. The parameter I is treated as a bifurcation parameter.

Table 2
Default parameter values used in the simulations of the BRC model (A.7)

BRC model parameters		
$a = 2$	$b = 5$	$p_x = 1.3$
$p_y = 0.4$	$k = 8.333$	$\varepsilon = 0.0012$

Introducing the slow timescale $\tau = \varepsilon t$ we obtain

$$\begin{aligned} \varepsilon \frac{d\theta}{d\tau} &= 1 - \cos\theta + A(x, y), \\ \frac{dx}{d\tau} &= k\left(\sin(p_x + \theta) - x\right), \\ \frac{dy}{d\tau} &= \sin(p_y + \theta) - y. \end{aligned} \tag{A.8}$$

Taking the singular limit, $\varepsilon \to 0$ of (A.7) yields the layer problem

$$\begin{aligned} \frac{d\theta}{dt} &= 1 - \cos\theta + A(x, y), \\ \frac{dx}{dt} &= 0, \\ \frac{dy}{dt} &= 0. \end{aligned} \tag{A.9}$$

Taking the singular limit, $\varepsilon \to 0$ of (A.8) yields the reduced problem

$$\begin{aligned} 0 &= 1 - \cos\theta + A(x, y), \\ \frac{dx}{d\tau} &= k\left(\sin(p_x + \theta) - x\right), \\ \frac{dy}{d\tau} &= \left(\sin(p_y + \theta) - y\right). \end{aligned} \tag{A.10}$$

The critical manifold for the BRC model is given by the set of equilibria of (A.9),

$$S := \{f(\theta, x, y) = 0\}$$

where

$$f(\theta, x, y) = 1 - \cos\theta + A(x, y).$$

Since $A(x, y)$ takes values in $(-1, 1)$, $\cos\theta$ must take values in $(0, 1)$. Thus, the critical manifold is defined when $\theta \in (-\pi/2 + 2n\pi, \pi/2 + 2n\pi)$ for $n \in \mathbb{N}$, and has a fold when $df/d\theta = 0$, i.e., $\theta = 0 \mod 2\pi$. Note that the slow variables are not influenced by the interval of θ in which the dynamics are evolving due to the dynamics being 2π-periodic. We may therefore consider the interval $\theta \in [-\pi, \pi]$, resetting the fast variable from π to $-\pi$ when the dynamics leave this interval. For $\theta > 0$, $df/d\theta > 0$ and so the critical manifold is unstable, whereas for $\theta < 0$, $df/d\theta < 0$ and so the critical manifold is stable; see Figure 9.

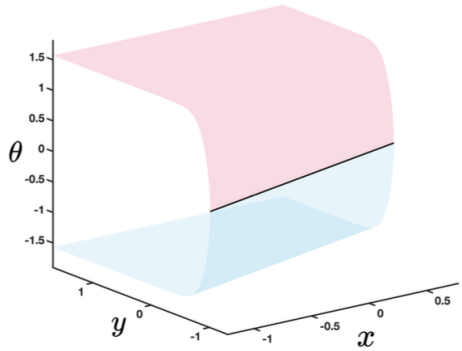


Figure 9. *The critical manifold for the BRC model, with parameter values as in Table 2. Stable equilibria of (A.9) are shown by the cyan-shaded sheet and unstable equilibria of (A.9) are shown by the pink-shaded sheet. Fold points relative to y are shown by the black line.*

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